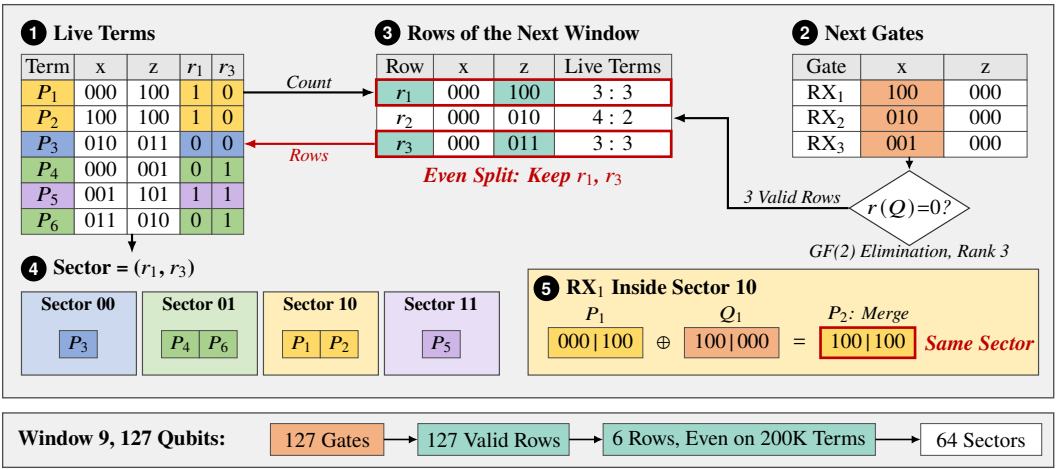

1 Live Terms
Term x z r_1 r_3
P_1 000 100 1 0
P_2 100 100 1 0
P_3 010 011 0 0
P_4 000 001 0 1
P_5 001 101 1 1
P_6 011 010 0 1
Count
Rows
3 Rows of the Next Window
Row x z Live Terms
r_1 000 100 3 : 3
r_2 000 010 4 : 2
r_3 000 011 3 : 3
Even Split: Keep r_1, r_3
2 Next Gates
Gate x z
RX_1 100 000
RX_2 010 000
RX_3 001 000
3 Valid Rows
$r(Q)=0?$
GF(2) Elimination, Rank 3
4 Sector = (r_1, r_3)
Sector 00
P_3
Sector 01
P_4 P_6
Sector 10
P_1 P_2
Sector 11
P_5
5 RX_1 Inside Sector 10
P_1
000|100
⊕
Q_1
100|000
=
P_2: Merge
100|100
Same Sector
Window 9, 127 Qubits:
127 Gates
127 Valid Rows
6 Rows, Even on 200K Terms
64 Sectors